

Loops Lab

1. Write a task that takes an array of 10 integers and returns sum and average.
2. Write a function that computes the factorial of a number N.
3. Given a logic [7:0] data, write a function that returns the reverse the reverse bit order of this vector .
4. Write a function to count number of 1's in a 32-bit vector.
5. Declare an array of multiple elements, fill with random values, find max/min using foreach.
6. Push 15 random values into a queue, then find how many are even, odd, negative.
7. Build an associative array int scores[string] with a few names, use foreach to print all key-value pairs and compute their total score.
8. Write sv code to generate a clock with 10ns period.
9. [**BONUS**] **Binary search** — Implement binary search algorithm on a sorted array, return index if target found or -1 otherwise.
10. Linear search in an array, break on match, return index if target found or -1 otherwise.
11. Write a function to compare DUT output array against a golden reference array element by element, reporting mismatches with index info.
12. Implement bubble sort algorithm on an array of 7 random integers. Hint : Nested Loops